

Active Inference LiveStream 057.1 ~ Active Data Selection and Information

Livestream | Active Data Selection, Information Seeking | 1:44:26

Hello and welcome. It's May 28th, 2024. We're in ACTIMF livestream 57.1. Welcome to the Active Inference Institute. We're a participatory online institute that is communicating, learning, and practicing applied active inference. This is a recorded and archived livestream. Please provide feedback so we can improve our work. All backgrounds and perspectives are welcome and will follow video etiquette for livestreams. Check out activeinference.org to learn more about livestreams and other projects. Okay, today in 57.1, we're continuing the discussion of the paper on active data selection and information seeking with Carl Friston and Thomas Parr joining. So, thank you both. We did 57.0 covering some of the background and to begin the 57.1, whoever would like to go first, how about Thomas? Lead us into the active data selection setting slash challenge and introduce yourself however you would like. Thank you and go for it. Okay, so I just introduced myself. I'm Thomas. I've been working in active inference for a few years and I split my time between research and clinical practice, which I think is part of the motivation for some of the later parts of this paper that relate to some of the clinical type problems that you might face with. I guess the background of this particular paper is that a couple of years ago, I was in Australia to present at the Australasian Basin Network Modelling Society, if I've got that right, I was in Australia to present at the University of Canada, who ran a really fascinating conference using Bayesian networks to model everything from volcanoes to coastal erosion and various other things. And there were quite a few clinicians involved in that conference as well in interest in things like respiratory diseases and diagnostics. And one of the organizers of that conference happened to be running a special issue on basic networks and invited us to take part in it. And so this was our contribution to labs. And the idea is to take some of the same principles that we use in active inference and some of the long-standing principles from people like our students like Lindley to apply to the problem of experimental design and to think about how active inference like principles can be applied outside of biology.

Sorry, I'm talking too fast for you to time.

Thank you. Okay. Karl, do you want to add anything to that or as we kind of begin?

No, no, no. I repeat, I'm an admiring spectator. This is

Okay. Okay. Okay.

Enjoy Thomas's lucid account of this very pertinent application of some basic principles of optimal Bayesian design and epistemic foraging. Okay, then no heckling. Okay, let's go to some of Chris's questions and also as people ask questions in the live chat, we'll go in there. Okay, so let's just start with time sampling. How did you or in what equations or in what conceptual ways did you bring in time and the kind of time sampling that was explored in this paper?

So time is a really interesting issue in terms of how you model a lot of these sorts of processes, partly because a lot of interesting processes evolve in time, but it also brings in some really

interesting issues in terms of how you sample and how you gather information about things. And the most obvious problem there is that some things change and some things don't. So that brings you into the world of a lot of what psychologists often deal with, things like inhibition of return, the idea that when you look at something, you learn something about it. The time it takes for it to be worth looking back there will depend on how unexpectedly things might change in that location. So that's a really important part of a lot of the sorts of generative models that we deal with both in biology and out of out of that area because you need to be able to deal with worlds that change.

And that sort of calls into question, how do you actually approach time in generative models?

There are a whole range of different approaches that one can take to start. So I think a lot of a lot of early work in active inference was formulated in terms of continuous time models, working with differential equations, generalised coordinates of motion, which is a way of representing trajectories in terms of positions, velocities, accelerations, as if you're doing a Taylor's expansion and taking the coefficients of that. And that would be one very good way of representing a temporal progression. A lot of the more recent work we do in active inference is based upon discrete time models, where you see things as being a sequence of events and you just represent what happens at a sequence of different time points. In this particular paper, we ended up using a slightly different approach and that was to essentially use a basis set or a set of basis functions to allow there to be some smooth progressions in time rather than dealing with specific time steps steps themselves. So it sort of blurs the gap between using discrete time steps where there's a clear boundary on the side of it and continuous progressions where there's no boundaries. By using a series of basis functions placed at different points in time, you end up getting this sort of smooth progression from one sort of blurry time step to the next. So I think time is sort of incredibly important in how we how we think about these sorts of things, again in terms of thinking about how things change, but also in terms of thinking what's the right way even of representing time in a generative model.

Awesome. Welcome, Chris, as well. This definitely is reminiscent of figure 4.3 from the textbook, which I'll bring up. How would you contrast it or which figure should we look at to explore which variables are continuous or discrete? This figure is the one here. So this paper won't have anything specific, but you're right, there probably is one from the textbook that deals with exactly this distinction. So here, I think one of the later figures does use a sort of basis set type representation to look at temporal progressions. I can't remember which figure that is, I'm afraid.

Okay.

Um, did you, did you want to talk through this figure that's up?

Yeah, maybe summarize this figure.

And so this figure was trying to, trying to demonstrate in a sort of relatively simple, um, analytic case, what it might mean, and sort of taking time out of it for the moment, what it might mean to select those data points that are going to be most informative about, um, some underlying model.

And the way it's been set up here is to start with, um, uh, so in the sort of upper left plot here, which is showing a function, um, with a whole series of different possible parameters that might exist.

So you can imagine these theta parameters that are sort of shown as one being bigger than the next, um, each represent which function might be used here.

Um, and, uh, and you can see that in some of those, there's a lot of variation in certain parts of the function. So, uh, the green one, you can see there's a sort of very big peak around zero, but slightly smaller peak for the purple one, slightly smaller for the yellow.

Um, but actually there is also certain points where, where it's not so obvious and you wouldn't really be able to distinguish it. So they all more or less cross the x-axis at the same point.

Um, and, and on both sides of zero. So if you were to choose somewhere on this plot to work out which function you were sampling from, you're much better off choosing from somewhere around zero, um, then you are choosing at the points where, where it's actually crossing the x-axis where the, where all look very, very similar. And so this is just to get across the intuition behind this process of selecting your data efficiently to be able to resolve between alternative hypotheses as best you can, or even to estimate along some continuous parameter value as it would be here. Um, so following the the sort of sequence round this, um, this graphic, um, the idea is that we can represent these things into several different kinds of generative models. So here we're representing either a Bayesian network where you say my explanatory variable is this theta that's generating some data y . Um, and then we have this π variable which like in a lot of active influence imitation represents our policy or our choice of where we want to sample. So here this is choosing somewhere along the x-axis that we might want to, um, work out what the corresponding y might be. Um, the, the factor graph representation deals with exactly the same model, but, but now expressed by putting the factors in explicitly. And by factors, I mean that our probability distribution, which is made up of our prior beliefs about theta and our likelihood, which maps from theta to our data y , um, are now sort of explicitly put in as squares that represent the different bits of that generative model. And the next step is to, to use a normal or a for a Forney factor graph, which essentially just omits the circles. So now the variables are represented by the edges, the points that connect the different factors. So you can imagine if we take our probability distributions, the probability of our latent states, our parameters, uh, and the probability of the data given that, and we separate those out into factors. And we draw a square for each of the factors of that probability distribution and connect those up whenever they share a variable. So the edges or the connections

between the factors are the random variables you might want to estimate. And then I'm trying to see behind the, um, cameras across. So the, um, the next step is, is to use the Forney factor graph because it's, it's

often quite a nice notation to use when trying to understand, um, message passing and the different things that

contribute to, um, to the way we estimate these, these variables. So here for each different square, we can see there being messages being passed between the neighboring edges. And when we combine those

edges, we can, we can arrive at a posterior estimate of the marginal of the variable we're dealing.

So this would be how we might go about estimating the theta there. But the key twist here is that we're now not just interested in message passing for the process of inference, but we can also now start to interpret those messages, um, in terms of how they contribute to various different entropies that together make our information gain. So here we have, um, both the conditional entropy, which we sometimes refer to as an ambiguity. And that conditional entropy essentially says, um, if I were to sample over here in a more biological setting, if I were to look over here, um, how confident am I in, in terms of what I would see or what, how confident am I that these parameters will back to something meaningful and informative? Um, the predictive entropy is another form of entropy or uncertainty that essentially says, before I've looked anywhere, regardless of these parameters, even if I've marginalized them out, how uncertain am I about the data I'm going to get? So the conditional entropy is how precise is the mapping or how imprecise is the mapping between parameters and data? Predictive entropy is just my uncertainty a priori about what I'd see if I, if I looked over there. Those things are quite interpretable when you start thinking in terms of experimental designs, because a, um, predictive entropy, if I'm already very certain about what I'd see or what, what would happen if I, uh, performed a particular experiment, there's almost no, no point in doing that experiment. There's only a point, uh, acting to find something out if I don't already know what I'm going to observe. So this predictive entropy, um, is a positive thing that if, if I'm uncertain about what I'd see, then I should go out and seek that after I should get those data points because there's something potentially to learn.

However, there are certain circumstances where, where all you get is noise and you may be completely uncertain about what you're going to get. You know, if you, if you just keep running a random number generator over and over again, you're never going to become much more certain about what you're going to see next. Uh, and so that's not a terribly useful thing to seek out. And that's where this conditional entropy or ambiguity comes in, in that it penalizes those things that are not going to resolve any uncertainty. So the way I often think about it is if you take predictive entropy, you have the total amount of uncertainty about what I'm going to observe. If you then subtract the conditional entropy from that, then you, you, you let with the amount of uncertainty that you might be able to resolve. And I think this comes back in a, in a bit more detail and we can discuss that a little bit more in some of the later figures if we get to those. Um, but essentially the difference between these two things is, uh, our expected information gain, uh, which is also a mutual information, which can often be expressed as the difference between those two entropy. How much am I going to update my beliefs? How much am I going to change what I, I believe as a consequence of making a particular

observation? Thank you, Thomas. That was a great summary. I think we'll return to this. I wanted to bring up figure 4.3 from the textbook. Ah, yes. Yeah. So this is, and yes, anything on this and to kind of juxtapose the conditional entropy with the A matrix and the sense making and the mapping and then the predictive entropy to the B matrix and the transition among latent states.

Yes, that's right. So, um, for those who haven't, haven't seen these before, these are again, sort of factor graph type representations where we're now dealing with, um, temporal progressions. Um, and so the,

the

graph at the top represents a series of states as they transition from one to the next over time.

There may be more or less uncertainty in those transitions. So I might be very confident as to how a sequence progresses, if there's very low volatility dynamics are very precise. Um, and then, and so those are the, the factors labeled three. Um, the factors labeled two represent the way in which my data are generated given some states of the world. So I might not be able to directly observe the S's, but they will give rise to the O's or the observable outcomes. Um, and exactly as Daniel says, there's a link between the, um,

the two different terms in, um, in this expected information game. So the conditional entropy will depend almost entirely upon this mapping from states to outcomes upon this, um, conditional entropy, um, will, will depend upon this likelihood distribution. Um, and so it will say that if I'm in this state, then there's a very precise mapping to the data. Whereas in, in alternative states, there might not be. And so that allows me to select between, between these things based upon, um, policy π I choose.

Um, the, the, sorry, the predictive entropy will actually be a function of both forms of uncertainty.

So uncertainty in the transitions. So if things change a lot since I last looked there, then my predictive entropy might be very high and I might want to look back there sooner. Um, again, in psychological terms, that's a, sorry, the lights have gone off. I'll see if I can turn those back on. Um, the, um, so the, the, um, predictive entropy deals with that sort of inhibition of return. Um, if it's due to the transition uncertainty, however, if it's due to the uncertainty in the, um, in the likelihood, then that will remain relatively fixed over time. And I won't be able to resolve that.

Sorry, let me see if I can sort out those lines.

Okay.

It's better.

So the, the, the, the, the, the, sort of lower part of the screen, so the, the, the lower graph is actually a similar representation, but now in continuous time. So here, when we're dealing with the, the states x might be continuous variables, the x prime is the, um, rate of change of my, um, my state x , the x double prime is the acceleration.

And if you know all of these things and further coordinates of motion, you can start to construct a trajectory or, or predict, anticipate both the future or the near future based upon these variables and the recent past as well.

Um, and exactly as Dan has written there, it's, it's effectively Taylor series expansion where these are the coefficients of that Taylor series.

So these would be sort of the two ends of the spectrum dealing with sort of a purely discrete time process and a continuous time process. And there are various ways of either blurring those boundaries or even if, um, even combining the two.

So maybe now having brought this out, could you return to the basis sets and comment on how it's similar or different to the time handling in the discrete time or the continuous time setting?

Yeah. So, um, so the use of basis sets will often imply that you're dealing with a continuous function,

but rather than have variables that represent either positions and velocities and those things, I mean, one could argue that actually a generalized coordinates of motion system is a form of basis set in itself. It's just a polynomial basis set. So you're, you're just wasting sums of different, um, orders in a polynomial series. Um, but there are other basis sets that, that one could use. And those include things like, um, uh, Fourier basis sets or cosine basis sets, where it'd be difficult to sort of localize things in a specific point in time, but as much frequency based, um, or as, as happens to be done in a particular paper, um, it, it's a series of Gaussian basis functions. Um, and those are centered on different points in time, but because they're Gaussian shaped, the boundaries between them are a little bit blurred. So the value at any particular point, how much you weight that particular function localized around a specific time point will also have an effect on the neighboring time points. And that's effectively a way of putting in a prior belief that, that, that although there's a temporal progression and perhaps a series of discrete steps, there's actually a smoothness to that as well. There are no discrete jumps in this that actually I'm just adding, subtracting functions that vary smoothly, um, to construct that overall picture. And that gives you quite a flexible way of representing a lot of, a lot of often quite non-linear functions. I think from memory that, um, I'll actually use a sort of basis set type functions, um, in some of the COVID models as well, and some of the later papers to model things like seasonal fluctuations, um, and changes in, um, transmissibility at the time. So they have quite a, there's quite a broad, um, way of using them. The, the downside of using a lot of those functions that they're, they're often less mechanistically interpretable. So that's something that, that are often quite convenient in terms of getting a flexible function representation, but wouldn't necessarily speak to a specific mechanism. Yeah, thank you, Thomas. Yeah, also basis sets are brought up in SPM, and I think one advantage that's raised there is how it allows the pre-specification of all of the oscillatory patterns of a given frequency, range, and then filtering the data with that pre-specified sieve in a way, um, interesting and compressing information about the hypothesis of what structure the data could have, like nearby things being more related, events in time being more related if they're close and so on. Yes, and compression is a good point there, that actually as soon as you start using a basis set, you've now got the option to stop after a certain number of functions. So in generalized coordinates of motion, there's a limit to how many additional coordinates it's normally useful to put in, because the degree of accumulation of the uncertainty means that once you're beyond a certain point, it becomes so uncertain that it doesn't necessarily contribute to the earlier terms of the sequence. And that sort of arises as a function of autocorrelations over time. And you're right to say that using things like frequency-based or Fourier-type basis sets also gives you that opportunity for compression. And I think a lot of image processing does use things like the discrete cosine transform, which is based upon a series of cosine basis sets where you can throw away certain orders of that or certain frequency bands without losing too much information. Awesome. Okay, let's go to a second question by Chris, but Carl or Chris, just raise your hand or totally go for it. Let's go to the second one. Let's ask this question by Upcycle Club. Do you think that there

might be any algorithmic modifications, meta heuristics or hybrid approaches that could potentially lead to even better data selection optimization?

Probably. I suppose it's a matter of thinking about what hybrid approaches is. It's a question of what what it's being hybrid with. And I imagine we may come on to some of those questions later on as well. So, meta heuristics. Well, I suppose a heuristic from a basic perspective is just a prior. And so I think it's exactly right to say that with the right priors in place for the right sorts of models and the right right sorts of settings, this process can be made much more efficient. So if you know a reasonable amount about the structure of your problem, then you don't necessarily need to rely on basis sets and that sort of thing. But actually, often you can write down something with a good deal of information into already. And then your model doesn't necessarily need to learn all of that. It's just able to then use the resulting predictive entropy and conditional entropy to pin down the more specific things that you want it to now find out. So, yes, I'm sure that it's probably not going to be a one size fits all in many ways. The priors are going to be those that are most appropriate to the particular problem you're dealing with, the particular agent you have. But ultimately, they all, I think, come under the same underlying problem and the same underlying principle, which is that if you select things in order to optimize your model of the world, then it's about finding the best fit with your model of the world. And that fit would normally be quantified using something like a free energy or a marginal likelihood. So I think it all comes from the same principle, but if you're a different sort of creature or a different sort of agent, then the implicit model you have, the way in which that directs your exploration to find out more to improve your model fit can be very different.

Chris or Carl want to ask anything or bring up anything?

Carl Schaafnick, Ph.D.: Not quite yet. I'm kind of thinking through, you know, talking about the time series and that sort of thing. So I think we can kind of keep going on.

Chris or Carl Schaafnick, Ph.D.: Yeah, Carl?

Carl Schaafnick, Ph.D.: Sorry, I was just thinking about heuristics and what it means to people in a certain kind of economics, I think about gigarensa and the like, which I just read as priors, the right kind of priors that have a minimal complexity. But I was just toying my head, what is a meta heuristic? It's a heuristic next door to a heuristic. So Daniel or Thomas, can somebody define for me a hyper prior perhaps? Yes, that's what it must be. It must be a hyper prior. That's what I'd assume. Chris or Carl Schaafnick, Ph.D.: Okay, thank you.

Chris or Carl Schaafnick, Ph.D.: Although, you know, what even is a hyper prior? It's effectively just a prior, isn't it?

Chris or Carl Schaafnick, Ph.D.: Yeah, I suppose it's a prior in the context of a certain kind of hierarchical generative model, absolutely.

Also, there's a very practical element, I think, about chapter six of the textbook and the recipe for modeling. So maybe to kind of bring in this practical element in the modeling of these adaptive data sampling, what is the recipe for active inference generative modeling? Maybe even as you've learned or updated in the two years since chapter six?

Daniel Schaafnick, Ph.D.: I mean, I think the basic recipe is still the same. So, I think what we do is think about what sort of model that we're interested in and what sort of question that's supposed to address. And ultimately, when you're using modeling in the context of data, sorts of questions you might want to ask are things like, which of several alternative models is the best explanation? So, doing a sort of discrete hypothesis test or trying to actually recover particular parameter estimates. So, coming up with posterior distributions over in active inference context, that will be over particular kinds of beliefs that then determine behavior. And that's particularly important in the context of things like computational phenotyping in clinical populations. So, trying to understand which beliefs might be important. And by beliefs, I mean sort of implicit Bayesian beliefs, not necessarily something explicit that you might be able to articulate. But what sort of beliefs might vary between different populations that then lead to particular kinds of what we might think of as pathological behavior. And so, you can start then ask questions in terms of, if you can phenotype different people, you can start to use those beliefs almost as if they were data that you can then ask other questions of. So, how does this vary with a particular diagnosis of therapeutic intervention? So, there was a figure in the book, which I can't remember the sequence of now, setting out the sort of sequence that you might follow if you were trying to develop a model and then use that to actually answer a question based upon some empirical data. But I think the key idea is, you're going to bring that up. I think the key idea was, if you have your model to predict some behavior from it, you can then use that model of that is effectively someone's brain's model of the world. But then you're modeling someone else modeling the world.

And here we are. So, I think in the...

I think this one.

9.1.

Well, so the figure you've sort of copied into the lower left there is quite a nice way of thinking about this process and active inference of modeling other people or things doing modeling.

So, someone's brain that you're trying to explain is itself modeling the world. So, from that brain's perspective, the data that needs to be explained are sensory data. So, visual data, somatosensory data, auditory data, whatever. And then, as a scientist, you're trying to explain the way in which that brain is behaving and generating behavioral outputs or electrophysiological signals or whatever you might be measuring as a neuroscientist. And so, my model of this brain's model ends up becoming the key thing. And that's sort of sketched out in this particular figure.

And the other figure that you had up for a moment, sort of talking through the sequence that one might adopt in terms of how you might construct this. And it's not necessarily linear, I think, as the arrows show. But essentially, you need to think about what sort of task you want to form or what experiment you want to form.

You need to think about what model that's optimizing. And here it's showing the partially observed mark of

decision process model, which might, in this case, be the internal model of, as shown here, a rat in a maze.

From that, you then need to construct a likelihood function, which says, given that the rat had this sort of model and these parameters in it, what's the likelihood that it would behave in a particular way?

So that P of U is the probability that they would have behaved in a particular way, given that model.

And then we might equip that likelihood of some prior beliefs about parameters in that rat's model.

And combining the prior and the likelihood, we can then invert that model in much the same way we do any other sort of Bayesian inference.

And there are further analyses we could then do on that.

So this is what I was sort of sketching out earlier on, that if you have parameter estimates and you know the posteriors for a number of different rats in a maze or patient populations or even artificial systems of some sort,

you can then perform other analyses on that almost as if those posteriors were data points themselves.

And you can start to ask, how do these beliefs vary as a function of diagnosis or experimental condition or whatever else?

Thank you.

Okay.

Let's...

In the...

In these beginning parts,

let's focus on what the model is and then as we kind of move towards the end and the second time talking, we'll look more at the clinical trial and maybe the questions with...

that are more open for the future.

How about connection with machine learning?

So...

Chris wrote...

Um...

Your examples...

Primarily focus on statistical modeling.

How do you see active data selection methods integrating with other machine learning models, especially deep learning architectures that typically require large data sets?

Are there specific challenges or considerations in this context?

I mean, I would argue that almost in a sense, the answers in the question that these large data sets that they often seem to require is exactly the opposite sort of principle as to what we'd be working with in this sort of paper.

So the idea is, how do I work with more efficient data sets?

How do I design not just models, but also the selection of data to make sure that even with a smaller data set, it can be more informative?

So I think there's a bit of a difference in the underlying philosophy, but I would also say that most machine learning models are forms of statistical modeling in some sense, whether or not they explicitly represent the

uncertainty that they get out of them.

I mean, I think it is also where, you know, the comment about hybrid approaches that was discussed in one of the previous questions as well.

One way of thinking about that might be in terms of using things like neural network architectures and function approximators as part of the, as part of a basing or statistical model.

And that's sometimes referred to as a process of amortization.

You can fix certain parts of your function approximation.

And in the sense, using the basis sets is also a form of function approximation that then simplifies, or in some ways makes it cheaper to do the process.

Next time you need to do that inference that you can sort of save a function that says, given this input, this is the outcome I should end up with, or this is the inference I should arrive at.

One of the challenges there is being able to go in the other direction and to be able to say, if I'm going to actively select data, then I need to be in a position where I can predict or generate the outcomes I might get.

And that might be something that is possible to do in the context of generative approaches in gene learning.

And I suppose all of the approaches we've been discussing really are generative approaches.

But because I can now generate not just the outcome itself, but I can generate a distribution over the outcome conditioned upon what I might do.

That's one of the key things in being able to, being able to select those data appropriately.

And if you're using a model where it's very difficult to project that into that space and to ask, what are these conditional entropies?

Then it does become very difficult to select data appropriately. So using something that's sort of very highly nonlinear and difficult to be able to work with those things, it does become more challenging to do that.

Sometimes people might sort of approach it using sort of sampling based methods to come up with an empirical distribution.

But by the time you start doing that, you end up with something that is much more costly.

So if you're if the problem you're working with is one way you want to try and bring it down to choose fewer data points and select your data more intelligently, then often that does require particular kinds of model that let you do that in a very efficient way.

There's an interesting area and it'd be interesting to see how, how, how you might attract these sorts of approaches.

One thing you might imagine is that actually if you've got a very clear, unambiguous mapping that you can learn with the deep learning type model, it might be that you could actually just treat the outcome or the labels that they should produce by that deep learning model as data from the perspective of another model might go about selecting the right sorts of labels that are going to be helpful in terms of learning and how things.

So there may be that you might have to look at the right sorts of methods.

So there may be levels of the model that can be when they're sufficiently unambiguous, sufficiently precise mapping, where you might not need to go to level of detail and select things in that way.

But it may be that the outputs of that are things that need to be combined to form interesting mechanistic hypotheses about what happens next.

That's very interesting, like thinking about different kinds of communications and reliabilities of different kinds

of analyses or computer systems.

And then it's actually the active data sampling conditional entropy in practice, which is that reliable sources are pursued as part of the information foraging adaptive ploy that every agent has to be involved in.

Like reliability is its own suitability for an information resource.

And then the other side of it is actually resolving the variability of the latent states.

However, those may have extremely high fundamental variability.

So often pursuing the conditional entropy, the agent has more control over because they can resolve something more.

At least they can resolve that.

Yes, it's knowing what is and isn't resolve.

I think that that's key because if you were to just seek out things that were very uncertain, you might be sat there pressing a random number generator over and over again without being able to learn anything from it.

And it's an attractive state that you end up being stuck in unless you can contextualize that with how resolveable it is.

I think one of the simulations we put together in this particular paper explicitly varied the uncertainty as a function of where you are along a particular number line.

And that was just to illustrate the point that actually there is a...

So it was before this section.

Yes, it would have been this one.

That's right.

So here you can see that we're dealing with a function where the variance grows as you move away from about X is zero.

And if you know that, then you know that actually the informativeness of the data at the peripheries is much less because you're not going to be able to resolve a lot of that uncertainty no matter how much you look there.

So if you see the choices plot in the lower right here, the solution it arrives at looks like a very sensible deliberate solution, which is to start in the middle where things are most precise, then fan outwards, moving left and right to resolve as much uncertainty as it can before it's then resolved enough that the cost of sampling further data is more than the amount of uncertainty that might be resolved before.

Whereas if the ambiguity of the conditional entropy were consistent across the entire line, then you might expect a much more even pattern of sampling and something that in practice might not be very different from random sampling, except for the sort of sequential effects.

I'd love to hear a first pass on how this figure addresses some of the similarities and differences between active data sampling and maximum entropy sampling.

Yes.

Yes.

So I don't know whether you have one of the graphics of one of the previous figures where it doesn't have this ambiguity to happen.

But so perfect.

Perfect.

Yeah.

So this figure shows an example where the ambiguity is actually the same everywhere.

So the way in which the data are generated, and you can see that because you can see that in the sample data plot, there is a sort of well-defined function.

But the amount of variance around that function is more or less constant throughout that entire line throughout that function.

So in this circumstance, the conditional entropy or ambiguity is the same everywhere.

That means that the only thing that really differs when you're deciding where to look next and where to sample next is actually the predictive entropy.

And so here it will just choose locations about which it's most uncertain.

And that's quite a sensible thing to do because there's nowhere that's limited any more than anywhere else in terms of the amount of resolvable uncertainty.

And so you can see that the prediction it arrives at after making a series of choices and you also see the series of choices it makes are fairly well distributed.

They're quite evenly spaced and cover a lot of that ground.

And so the prediction it comes to actually matches that uncertainty quite well in a maximum entropy.

So this is this is the identical in this particular setting, the maximum entropy approach is selecting those locations that have the maximum predictive entropy.

That's very different to what we were seeing on the previous figure that Daniel was showing, the one where where it starts in the center and fans outwards.

Because if you were to adopt a maximum entropy approach here, then then you can see that you'd spend much more time sampling in the far peripheries, which is actually not a terribly useful thing to do here because because you're going to just get a lot of uncertain data.

You're not going to be able to resolve it very effectively.

And so here, the big difference is that ambiguity version forces you to start in the least ambiguous places.

And only once you've got the most information you can after that, do you start planning outwards and going to the more uncertain regions either side, where there's actually less information overall you can gain because most of the uncertainty is now unresolved.

The other thing that we've done with this plot and I think the other one as well is specify a cost to sampling cost to choosing the next data point.

And this is very much like we do in active inference simulations where we might trade off exploration and exploitation, so trade off having preferences for one thing over another.

And this is just to get across that idea that there often are going to be energetic or computational or even financial costs to going out and getting more data.

And those costs are specified just as prior beliefs about what I will tend to do in the absence of any potential information that I might gain.

So being a prior belief that can just be added to the log probability of me stopping sampling can just be added to the mutual information or expected information gained, which then gives you an automatic trade off between

exploitation and exploration.

And effectively means that if I keep exploring, here we are exactly, so this C here.

So once the expected information gained, once the amount of uncertainty I can resolve drops below the probability that I would stop sampling, then I will effectively just stop sampling.

So you get an automatic cutoff as to when you want to stop doing your experiments.

That's a very interesting piece.

This was kind of a cool aspect of it.

And thank you for that connection to the maximum entry.

Chris, want to ask anything or we can ask a question from live chat or one of your prior questions?

Not sure it's necessarily a question, but more I appreciate your answer for that.

That last question that we were just discussing because it makes a lot of sense and it almost seems like at some point in more complex models that you can't really necessarily know exactly what you're feeding.

Some sort of machine learning or neural net, you might want to take a hybrid approach where you have something that you select a small amount, select down your sample, your data down to a certain size, and then you can kind of tweak and optimize your model a little bit more by throwing a couple of random things in there.

But at least you have a smaller set that you're starting from.

I mean, is that something that you can kind of see doing is like instead of trying to sample down to something that's most informative sample out what is the least informative?

Yes, I suppose that's the same sort of approach, isn't it?

I mean, you can't see this starting in the least ambiguous place, but the low hanging fruit and then moving out from there until you feel you're not really gaining anything further.

And I think you're absolutely right.

This is as long as you're able to specify the cost of carrying on sampling and seeking out new data, that gives you quite a principled way of working out when you stop doing that.

Because a pure maximum entry approach will carry on going because there's lots more entry, there's lots more uncertainty.

And so you might want to resolve that unless there is cost to doing so.

And it's a perfectly reasonable thing to do it. There is no cost to doing so.

You can just keep looking.

But as soon as you're dealing in a realistic setting where actually by doing this, you're not doing something else or you're investing some resource, be that energetic, computational, financially, then taking account of that can be quite a useful way of doing that.

I mean, it's an interesting point about more complex and large models.

And I suppose one of the key points there is that the best models are always going to be the simplest models you can get away with.

So if you have very large models with lots of parameters, the advantage is they're very expressive.

You can fit across lots and lots of functions or different patterns of data.

The downside is, first of all, that they become less interpretable.

But second of all, that you expose yourself to problems of overfitting.

And one of the key things in the active inference approach is that by minimizing free energy or marginal

likelihoods, you're implicitly minimizing complexity.

those data that are literally good because they are informative, that will technically or operationally increase the mutual information of inherent in your model between cause and consequence, so I think that you know there's something really quite fundamental about this sort of whole notion of being active in the way that you deal with data and gather information that you know that underwrites active inference as you know with its special focus on active sensing and active vision and active inference and active learning that you can actually go right through to active natural selection you know and one has to wonder now whether there's a homologue in evolutionary thinking of this kind of active learning that you know. Thanks for those points, those are great. Yeah, Thomas, go for it.

I don't think I had anything really to add to that, it's exactly the, it's being able to decide also upon your model space, isn't it? I think it's part of the point that Carl's making there that it's actually it's the selecting your model but it's also knowing which models even to select between is another key part of it.

And that's certainly been an interesting area in terms of structure learning, is that to interest literature.

Absolutely. I really like the call to evolutionary biology, I think that there may be an analog there, you can kind of wrap your mind around some of the things that you're doing if you frame it kind of from an evolutionary standpoint too. It's alternative way of looking at it but I think it's quite valid.

Yeah, also, those two points there, they speak to a key issue in structure learning, read as basic model selection, do you take a top-down approach or do you take a bottom-up? Do you grow your

models carefully by accepting this mutation or this extra state in the factor or indeed an extra level in a deep generative model in the face of this data or this evidence for how good this model is, you know, almost in the sense of adaptive fitness in an evolutionary context or do you use good old Bayesian model reduction and start with something that can do everything and then prune away in the bits that you don't want. And just thinking from a biomimetic point of view or an evolutionary slash developmental, at least neurodevelopmental perspective, sort of evo-devo, it looks as though biology does both, doesn't it? So we start off with a brain which is over-parameterized, over-wired, over-connected and then carefully prune away using Bayesian model reduction what we don't want. And yet, of course, at the level of exploring that model hypothesis space, I would imagine we rely a lot on carefully crafted mutations split and merge like operators that, you know, gently feel out into different, you know, from that perspective each of us is a data point for evolution. That's quite chilling, isn't it?

A lot to say on this. One really practical note with the cost of sampling and the opportunity to stop is to have different information foraging locations that have different costs, like different APIs you could call or different costs associated with taking on or holding a record. So it's really true, what multiple of you have alluded to, like it does reframe the question into creating this semantic kernel of understanding, for example, the clinical trial setting, something that can be shown up on a graphical representation and then being able to use that purely in the generative forward direction to generate synthetic data with whatever other evaluative criteria applied to that or just to learn active inference.

And then also to use that generativity, not just to make more data, which is often of limited utility for retraining,

but actually includes that adjacency of being able to ask how much another piece of information or evaluating the last incoming piece of information or rank the informativeness.

However, in a way it moves all the challenge into how you define the generative model, surprise, surprise, because you could have a model of information in the world that would lead to various attractor states, repetitive looking, fixed looking, and so on.

So it's pretty interesting how I guess that's the high road and the low road kind of coming to meet each other in the information foraging setting.

Because the fundamental constraints and trade-offs associated with sampling, they do come into play and it does not solve the question.

It just sets up some of the objective functions that can be used and have interpretability for these uncertainty situations.

Okay, I'll ask another question from the live chat.

All right.

All right.

Dimitrios asks, what is the nature of the precision weighting mechanism in this process?

Is it possible that high precision of the prior blocks the prediction error to convey the newsworthy information?

So precision and the balance between priors likely to do

and what determines which precision we might use.

And I suppose relating that explicitly back to this question of information seeking.

So precision can be applied to any probability distribution essentially.

So you could have a precise trial, you could have a precise likelihood

or imprecise versions of either of those.

And this gets particularly interesting and sometimes quite confusing in the context of hierarchical models where the prior from one level is effectively the likelihood or conditional distribution from the perspective of the level above.

And so you end up with questions, am I increasing prior or am I decreasing the effect of the likelihood?

And it often depends upon exactly where you are in these models.

But taken to its simplest level, you imagine you have a prior and likelihood

and the data that you are evaluating right to it in relation to.

The relative precision, and it is the relative precision that's important between the prior and likelihood will determine which one of those has more of an important final beliefs on your posterior.

In a very precise prior it says, I'm so confident about this that no matter what I see, I'm not going to shift my beliefs.

And relating this back to the concept of information seeking, that would be the situation where where we're in a dynamic setting.

I think there's been so little change from the last time I saw something that even if I saw something new

it wouldn't make any difference.

The likelihood, if that is very precise, leads to very low ambiguity.

So if I believe there are very precise regions of space from the perspective of my likelihood or regions of whatever process I'm selecting between the different policies I could adopt, then I'm more likely to select those locations compared to others.

And this is often in the context of neurobiology and psychology, these are often how people talk about processes like attention.

So being able to say that my likelihood is very precise in the context of spatial attention associated with these particular spatial channels or this area of space, then that means I'm going to update my beliefs much more based upon this thing that I'm paying attention to.

The link back to information seeking is that because I might update my beliefs much more based upon this location in space, I might act in such a way that I get more information from it.

I might move my eyes to look over there or take whichever action is relevant to whichever data set I'm dealing with.

And we can think of that as essentially saying this region of space, this region of policy space is more salient than others because when I choose to do something that leads to that outcome, I'm going to update my beliefs much more as a consequence.

So I think that sort of gives you a brief summary of how priors and likelihoods interact with one another.

The notion of prediction error came up as well, which I think is an important one to think about, particularly when you're dealing with continuous states based models and perhaps most simply when it's a simple linear Gaussian model.

Because the degree of update that you might make is effectively going to be a linear error function, where that error is the difference between what I predicted and what actually happened.

And that weighting is going to be weighted by the precision of the likelihood.

So the degree of update I'm going to make to my beliefs depends both on the precision, how unambiguous I think it is, and on the degree of, on the amount of discrepancy from what I'd originally predicted.

And that's the same idea that then underwrites a lot of predictive coding and a lot of neurobiological models.

Of course, even that needs to be weighted very carefully by the prior precision and the equivalent errors that you get once you deviate from your priors.

And that probably brings us back to this notion of complexity and the need to keep things simple.

Because one reading of keeping your model simple is that you keep things as close as possible to your priors.

If you have lots of parameters and they're all deviating from their priors quite a lot in order to come up with an explanation,

then that's effectively a very complex model, one that I think would be penalized relative to a simpler model, where everything stays much closer to the priors you have.

Thanks. Let's go a different direction.

There was a very interesting and ambiguous use of the streetlight effect.

So what is the streetlight effect and how is it kind of a representative setting?

And what do we do?

I love the graphics.

Oh, there were so many fun, fun representations.

And so what, what, what is it?

Is it, is it, is it a joke?

How did you find, how did you find one with a pie?

Oh, just typing in different symbols to see underneath the spotlight.

Very good.

Yeah, GPT-4O.

So the, the idea of the streetlight effect comes back to this idea of ambiguity, aversion.

And it's an idea that, I can't remember where it first came from,

but I first came across it in psychology literature.

I think possibly a reviewer mentioned it in a paper at some stage.

But the, the streetlight effect is originally set up to be something to describe an unhelpful bias.

And the idea was that if you're sort of coming home late at night and it's a dark street and you drop your keys, the first place you might look is underneath the streetlight.

And that's unhelpful in the original reading because your keys are not more likely to go under the streetlight than they are anywhere else.

And so it's a metaphor for seeking out those sources of information that are most easy to, to, to acquire and, and that you can learn the most from very quickly.

Of course, from an active inference perspective, that's actually a completely optimal thing to do.

That if you have some uncertainty to resolve, the very first thing you should do is look under the streetlight, because then you'll, you'll quickly establish that your car keys or your house keys are not there.

And then you might start looking elsewhere.

So another way of putting that is that the least ambiguous place, the place where there's the clearest mapping between the location of my keys and the visual data I would get is under that streetlight.

So if I'm ambiguity averse, if I choose those places where that conditional entropy is lowest, then I'll start at the space under the streetlight.

And maybe I'll then fan out from there, just as we saw in the simulation results that you put up earlier on.

Yes, what a, what a comical plot twist.

That it's like, it's a joke because it brings us to the point of laughing at the person looking under the streetlight.

And then it leaves it open like, but what should they have done?

And then this isn't exactly what they should have done in this paper.

But just like you said, it's a reasonable starting location, especially if it's where one finds one to be.

So it's kind of interesting.

Um, Carl or Chris?

No, I love that.

Yeah.

Yeah.

And I haven't heard it phrased like that before.

It's very annoying.

Chris?

Yeah.

Nothing really.

I do like that analogy though.

Let's.

How was equation three and this seemingly very general.

usage of, of information gain on, uh, base graphs.

How did that come into play?

And maybe if you could just define how that was used.

So the idea here was that, that just as in, um, a lot of Bayesian inference problems, where you're dealing with relatively complex generation models, um,

appropriately complicated,

appropriately complicated,

to help accurately explain data.

Um,

there, there's often quite a lot of sparsity in those models.

There might be multiple different, different nodes and edges, uh, that one might need to take advantage of, uh, or sorry, that, that represent variables one might want to draw inferences about.

Um, and so there's a long, um, and so there's a long, um, and so there's a long, um, sort of series of, uh, papers and advances that have been looking at ways to do that efficiently from an inference perspective.

Uh, and that, that has led to many, many forms of message passing algorithm, um, relying upon different, um, approximations to, to posterior probabilities.

Those include, um, sort of, um, trying to remember what the very early ones, or I think the Baum-Welchard, uh, algorithm and things like that were sort of early versions of, of what then ended up becoming fully Bayesian type approaches.

And the more general forms would be things like variational message passing, which relies upon field approximation.

So the idea that I can factorize all my variables or the posterior probability over all the different variables I'm dealing with.

And by not having to deal with all the posterior covariances and conditional dependencies, I can come up with a much more efficient, but still relatively good and useful, uh, set of beliefs about, um, about the causes of my data.

Um, then there are more, um, more accurate algorithms that were then, uh, developed as well.

So things like belief propagation, some, uh, some product type algorithms, um, which implicitly do take

account of pairwise dependencies, uh, between different variables, um, but still allow you to come up with good marginal inferences.

Uh, often these rely upon subtly different free energy functionals.

So for instance, belief propagation implicitly comes from something known as the better free energy, um, which is B-E-T-H-E.

Um, and that, that effectively assumes that the posterior factorizes in such a way that, that I can take account of all the pairwise, uh, dependencies and then renormalize by all the singleton.

So the distributions of each single margin in my system.

And that allows you to, to achieve exact inference from the perspective of the marginal distributions you get out of the end.

The point of doing all of these sorts of things, uh, and, you know, there are many others, expectation propagation and various others sort of based upon other free energy approximations.

Um, and, um, one of the general is, is, uh, a Kikuchi free energy that can be factorized again in different ways.

But the, the purpose and the advantage of doing all of these things is that it allows us to, um, to make use of the conditional dependency structures, the sparsity structures, um, and,

um, and, listen, the mark off blankets in a, um, in a network and, um, ask questions about specific variables.

Solve that in quite an efficient way to be able to say what are the beliefs we have about.

The idea behind the way this equation was phrased is that you can actually use those same messages in the same person structures where, um,

beliefs depend upon the other beliefs in the network and the message passing the results of that.

So, um, to work out the information gain about specific variables.

So you can, you can make use of exactly the same, um, technology that was used in things like belief propagation schemes,

and which underwrite a lot of neurobiological message passing schemes.

So this message, sorry, this, uh, equation, um, uses the notation CH to mean children.

So the things upon which, um, the things that depend upon the variable we're interested in,

uh, the parents, um, which are the things that it depends upon.

Um, and those things and, and, and the parents of the children, which you'll see also in a, in a couple of locations here,

which, um, together, so the children, the parents, and the parent, the co-parents of the same children,

together make up the Markov blanket, which is all the things that, um, that if I knew about,

would give me all the information I needed about the thing I'm trying to infer.

So I only need to know about local blanket surrounding a particular node and not the rest of the network, because if I know about those things, that tells me everything I need to know.

And so this is just a simple way then of, of, um, taking advantage of, again, that sparsity,

and to be able to come up with an efficient way of estimating not just the posteriors,

but the information gain I could potentially achieve about those posteriors given a particular policy.

Or the information I could gain about those variables with posterior relative to the prior rate.

So the, in the textbook, there's an example where the generative process and the generative model have the same form.

Like if they're both generating from a Gaussian, then those distributions can truly lock.

But also it's possible to have the form of the generative process and the generative model differ.

So like the generative model is doing some categorical, too cold, just right, too hot, and then actually temperature is a continuous variable and so on.

So how, how does the selection of family of distribution that's even used relate to the information gain like in table one?

Okay, I think, um, I'll just try and unpick what I think, but possibly two different things there.

So what one was this notion of whether the generative model that we might use our prior and like could actually match the way in which the world around us generates those data.

And I think that's a really important thing to consider.

And that exactly, as you say, there are often situations where the best model is in a sort of lay sense, the wrong model,

in that it doesn't match how data are actually generated, but is of simpler explanation that it complies with Occup's razor.

And the reason for that is you can always generate the same data in a more complex way.

So if you can get down to the simplest way of generating it, that's often the best way.

One of my favorite examples of this was something I found out a while back when we were doing some work on modeling of eye movements.

And one of the things I found is that it's much simpler to model your eyes moving together as if you just had one eye,

and then make predictions from the perspective of both eyes, because both eyes move together.

Excluding sort of subtle things like convergence.

Most of the time when you want to decide upon a gaze direction, you don't need to decide on separate gaze directions for each eye.

You might if you're a chameleon or another sort of creature that can move their eyes independently.

But from our perspective, actually, you can make do with a generative model that's much simpler than the physics of your eyes,

where actually there's nothing to say that they couldn't rotate two different spheres in completely different directions.

So by doing that, you actually end up with a model that is wrong, but is also a much better model than one that includes lots of additional parametrization and degrees of freedom that are just never used in reality.

So I think that sort of is quite a nice intuitive example of where your model might differ from that in the world.

And then perhaps related to that is the other point you're making about whether the particular distributions in your model are consistent with the distributions in the world.

Or even how do the particular kinds of distributions, forgetting whether or not they're accurate models of the world, how do they affect how you then seek information?

And it's of course there is.

So one of the reasons for including the table that you showed is that the form for the information gain will be

different depending upon which kind of model you have.

And that's just useful to know in the sense that if you're writing down a model, to know an efficient way of sampling from it.

It's going to depend upon what form that model has.

So one of the big distinctions that we often see and often make use of is this distinction between categorical models and continuous models.

So the two middle rows here are dealing with categorical or discrete priors.

And they have a particular form in terms of information gained and the way in which you do that updating.

Pull out of that the ambiguity bits and the predicted entropy bits.

The lowest row, the third row here, shows the form of a Gaussian model.

And one of the key things you can see in the Gaussian model is that the covariance predicted, so this σ^2 , is the predicted variance for the data.

And so that is going to scale with the ambiguity.

So you can immediately pull out from that how much ambiguity there's going to be just by looking at this term here, exactly what you're highlighting.

Now, if that depends upon π , then the ambiguity really matters.

If that's constant, as we saw in some of the earlier ones, and it doesn't depend upon π , then the only thing you need to take account of is actually the first bit, which is the maximum entropy bit of these equations.

So the form that you choose for your model is absolutely key in terms of how you then go about sampling the world.

And how you optimize your model is then absolutely dependent upon how you've sampled the world.

So you get this sort of nice circularity that then determines how you build your model based upon what you see.

But based upon how your model currently is, and what's uncertain about it and what there is to resolve, you're going to considerably change how you go about sampling it.

That's a really fascinating point.

And it makes me think about learning.

Like in a given situation, one could over or under learn or however it comes to be that their policies do matter for resolving uncertainty.

And then that could set up the balance between continuing to sample from distributions and or stopping them.

Yes, that's an interesting thought, because you could start with a model in which you don't know about how your policy affects, for example, this covariance.

But it may be that over time you realize that when I do this and end up in this sort of state, actually it does affect it.

Or when I take a particular action.

And so if you were then optimizing your model and then you go one step further and you say, well, how do I learn which or how do I infer which parts of my model will be affected by the actions I take?

And how do I choose the right actions so that I can best learn about which bits of my model will be affected?

And I think that's a really interesting question as to how you might approach that.

So we could have parameterized here or we could have included additional parameters that are how it does

the covariance depend upon policy.

is that we could make this is known to make this is known to make this is known to make this known to make this is known to make this known to make this known to make this

example if there is a value that can be placed on a given piece of information and that could be how much it influences some trade or it could take into account like the cost of paying the employee for that much time or using the computational resources then it's possible to allocate fixed or variable costs and stopping times completely separated from the epistemic and pragmatic generative model itself but with a cost as purely an exit boundary and not be not directly consuming the output of the free energy calculations

yes and that made me think of a couple of things so what what one is that that one of the advantages of that sort of cost is that in a dynamic environment it may be appropriate at some points to rest and then as uncertainty accumulates beyond a certain point you might want to go back to sampling so if you have a sort of time or cut off after which you just stop then then it's no good for anything that might have changed but if you if you've stopped for a little while and decided not to put in those costs you can then decide when's the appropriate place to start so again not just to stop the other thing i thought was interesting about about what you said is is again this notion that that to some extent although from an active interest perspective and certainly from the perspective of this paper information in itself is valuable and so the the value of information is in terms of how informative it is regardless of how it affects really what what might gain in the future however then how much you might gain in the future may also be relevant in terms of that pragmatic map and this brings in some of the ideas that have come through sophisticated inference like schemes um where where you start to evaluate these things recursively so in in in a lot of reinforcement learning um in probably most of it where there is information gain it's often um seen as the information that will lead me to the uh the result i want it will lead me to more reward uh and so creatures may learn a path through their environment that um that is the one that more reliably leads them to reward because it tells them where to go earlier on something those lines um however that one could say that that and that is important one one could also say that it's important to learn about your environment for its own sake as well and one of the things active inference does is it doesn't put any special um focus on either the fulfillment of your preferences or the seeking of information but the interesting connection between the two is that often seeking out information will help you fulfill your preferences in the future and so if you're able to recursively evaluate if i knew this what would i do uh and if i did that what would i learn and if i knew that and keep going into the future you can also select uh things that will give you information that will help you satisfy your your pragmatic value as well so information ends up having a dual purpose first of all in being valuable in its own sake but also being valuable in terms of what else you can gain from it in the future both in terms of information but also in terms of in terms of the um pragmatic value to be able to achieve

i think we'll we'll come to that with the um pragmatics of epistemic value but just one note on the message passing uh you mentioned how with the stopping criterion it's possible to think about the information foraging setting not just as like where to look every time a metronome clicks but about scheduling bouts like batching emails or something like that and that reminded me of our recent discussion with some of the rx infer team about the rx environment package which uses the reactive message

passing to uncouple the timing of for example the nestmate and the niche in a generative model so that they can be on different timings and the computations occur on demand like in a node local way and can have uncoupled timings so it's just very interesting how this pathway that is laid out in the paper about how on one hand we have the graphical semantic layout of what these variables are and then how many

preparation stages before it all gets played out as the relationship of the equations that we get from the entropies

so in a sense this is sort of setting out the process that someone might go through if they were trying to unpack this this sort of modeling in practice you might assume that actually a lot of it just starts from this message passing stage and the interaction directly with message passing and what you sample and a lot of the graphics along the top really to set out the equivalence between different sorts of formulas in which some people are very familiar with using one sort in principle you could tell this story without referring to the story without referring to a factograph and one of the big focuses really here was just trying to point out the way in which they're very related

to the competition

Carl or Christopher, we can go any direction.

I go in two directions because there are two really interesting issues there.

I can't resist, perhaps in reverse order, the notion of reactive message passing, I

think is not only crucial and again very biomimetic but also very relevant in terms of computer science and efficiency in terms of implementing any kind of active inference or learning scheme.

It actually inherits from something called the ACTOR model in computer science which is a big thing which led to the development of a whole suite of different programming languages.

One of the most popular nowadays is Rust that is predicated very much on this notion of responding to queries and requests for computation and the products of that computation which fits very comfortably with the maths of message passing on factor graphs.

I think the Rx infer reactive message passing is exactly the right way to go, not only in terms of understanding biomimetic dynamics and belief updating, but also in engineering the next generation of deep learning or artificial intelligence of a more generalized sort.

But the other direction I want to do is come back to this, I think, really important discussion about is the value of information intrinsic to the information gain or is it always conditioned upon your ultimate goal?

I think that's a really quite a profound point.

And I have a very specific take on that that inherits from the physics you get from the free energy principle.

So if you just write down what is the probability distribution over paths into the future,

and then you take the logarithm of that probability to create a potential that is expected free energy.

So the expected free energy that can be carved into the expected information gain and the expected log prior preference or expected value falls out of the physics of self-organization.

And that decomposition tells you that these things are linearly separable.

It means that there is no value information about the reward that does not exist because about the reward suggests you can't linearly separate the expected information gain from the expected value.

So this notion, I think that you only have to learn about that which you need to know in order to get your reward.

I think it's fundamentally wrong.

And one has to ask, where did it come from?

And I think it's by putting a certain kind of teleology and semantics on the physics of sentience and sentient behavior that leads to that false conclusion.

Because now what does value mean?

Oh, well, everything has to be in relation to the reward or the goal.

But of course, the value of information is not.

Another way of looking or if you like, dissolving that sort of disconnect between the intrinsic value of resolving uncertainty about the world as opposed to the extrinsic value of avoiding expected surprise.

Where surprise is read as a sort of negative log of your prior preferences.

It's just to think about, to acknowledge that we're talking about effectively Lagrangians or potential functions because they're log probabilities.

So when I say you can linearly separate them, what we are separating in a linear non-interacting way are the log probabilities.

When you actually combine the two probabilities, what that actually means is that I will only go for information gain if it satisfies my goals or equivalently and symmetrically.

I am only going to satisfy my goals if I get a sufficiently high information gain.

So they do contextualize each other in probability space, even though they don't interact in terms of information or in terms of potentials or log probabilities.

The final point is that you know this non-teleological conflation of the value of information and the value of conforming to the constraints supplied by the prior preferences.

Is that that is exactly what you get when acknowledging that the free energy principle is dual to James's principle of constrained maximum entropy.

So in this instance, you can just rewrite or reinterpret the free energy principle as an instance of constrained maximum entropy principle.

And you were talking before about the benefits of maximum entropy.

And this is one very simple instantiation of it where the entropy in question is the entropy of your posterior or your measurement, but it's constrained.

So now rewards are just ways of writing down constraints on an information maximizing system or a maximum entropy system.

And where do those constraints come from? They come from the generative model.

Where does the generative model, what constitutes generative model?

It's just the prior preferences in your generative model from the point of view of the free energy principle.

It's just a way of writing down a constraint.

But in log space, they're completely independent.

I think it's a really interesting point, which is, you know, confused me for years and, you know, also is a recurrent question when you talk to people in reinforcement learning, who have this, if you like, gamblers fallacy about the relationship between the two kinds of value.

And in practice, there's the parameterization of what the actual values are.

And that was something in the approach of this paper that Christopher and I had explored a little bit how it starts with the information gain, and then the end of the paper brings in the pragmatic value, as opposed to looking to expand the epistemic scope of reward and reinforcement and introduce some kind of variability or learning or intelligence layer.

So that's very interesting.

Chris, do you have any other thoughts or directions?

I just really appreciate how the paper's written, how well it has it, you know, mentioning what Daniel said, how it builds from a single kernel of information, mutual information gain, and then it just builds out a model from there

into something that is more and more applicable to different regimes.

So you can kind of stop at each point that you, any point, at any point that you want.

And then it just kind of grows into something that is, you know, immediately applicable to clinical trials, which is quite wonderful.

And I really did appreciate that.

I'm just going to have to sit through and probably process some of the things that we've discussed here today.

Yeah, I think that let's just in the last minutes think about what we might want to process over the next week and prepare some questions.

It might be cool to see the code.

So Thomas, if you want to have it loaded up, or we can try to prepare it on one of our machines, Christopher.

Yeah, or Thomas, what would you look forward to discussing or doing?

If you'd like to, it's all available on GitHub repository.

And I think it's set up so that if it's run, it will generate a series of animations showing process of sampling as well.

So if that's interesting.

Yeah, that would be awesome.

Yeah.

Let's return on that and look at the code.

What's the easiest thing?

Would you like me to send you something there?

If you could just prepare to share your screen with the code running, and then we can start with just a little bit of looking at it, just seeing what the code does.

And then in the second part, talk more about the clinical trial setting and how it is all brought together in Section 6.

Sounds good.

Cool.

Any last thoughts from any of you?

None here.

Thank you, Thomas and Carl.

This is awesome to have you join.

Thank you, Chris, for helping prepare this too.

So, all right.

See you next time.

Thank you, guys.

Have a good one.

Bye.

Bye.

Thank you.

Thank you.